

JACQUES NICO L. BELMONTE

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EDUCATION

Camarines Sur Polytechnic Colleges - Main Campus

Bachelor of Science in Computer Science

Nabua, Camarines Sur
August 2020 - July 2025

Awards/ Honors: 2nd place at GAWAD LIKHA CSPCeans Campus Research Fest | **GWA:** 1.5660

Related Coursework: Information Management, Artificial Intelligence, Machine Learning, Data Structures & Algorithm, Algorithms & Complexity, Operating Systems, Robotics, Application Development & Emerging Technologies, Information Assurance & Security, Software Engineering, Defensive Programming, Network Security, Digital Forensics, Human Computer Interaction, Automata Theory & Formal Language, Applied Cryptography, Platform Security, Modeling & Simulation, Calculus for Computer Science, Discrete Structures, Ordinary Differential Equations, Networks & Communications

WORK EXPERIENCE

Computer Science Research Consultant

Self-Employed

Nabua, Camarines Sur
April 2026 - Present

- Guided students in checking their manuscript, designing and implementing Abstract Syntax Tree (AST) parsing systems using tools like JavaParser/ANTLR to analyze the structural composition of Java programs.
- Mentored the development of structural similarity detection models using tree edit distance and graph-based techniques, enabling accurate detection of code plagiarism beyond surface-level text comparison.
- Supervised experimental design and evaluation of systems using accuracy, precision, and recall metrics, ensuring research outputs meet academic and technical standards.

Part Time Graphic Designer - Virtual Marketing Assistant

Doneverse

Nabua, Camarines Sur
September 2025 - December 2025

- Designed and delivered high-quality marketing visuals using Canva and Adobe Photoshop, supporting branding and digital campaigns for multiple clients.
- Implemented marketing automation workflows using Go High Level, successfully assisting 2 clients in streamlining lead generation, follow-ups, and customer engagement processes.

Freelance AI Developer

Self-Employed

Nabua, Camarines Sur
July 2025 - Present

- Successfully delivered 2 client-based AI applications, leveraging computer vision solutions to solve real-world problems, from development to deployment with fully functional outputs.
- Utilized Python with libraries and tools including YOLOv8, YOLOv9, OpenCV, Streamlit, Pandas, NumPy, Matplotlib, and Seaborn to perform computer vision tasks and exploratory data analysis (EDA), building interactive and data-driven applications.

Cybersecurity Compliance Unit Intern | Python Developer Intern

Directorate for Information and Communication Technology Management - Philippine National Police Main Headquarters

Quezon City, Metro Manila
March 2025 - April 2025

- Completed 162-hour internship in the Cybersecurity Compliance Unit, contributing to the development of a confidential phishing simulation campaign and supporting secure system practices within the organization.
- Developed Python-based security tools and assisted in drafting cybersecurity policies and digital safety protocols, utilizing libraries such as cryptography, hashlib, hmac, ssl, smtplib, email, requests, and os for secure data handling, encryption, and automation.

PROJECTS

Real-Time Drowning Detection using YOLOv8 | Python, HuggingFace, streamlit, opencv, RTSP, ultralytics, PyTorch, torchvision, NVIDIA CUDA, TorchVision, YOLOv8, Tapo, Python, TensorFlow, Pandas, NumPy, Scikit-Learn, Pandas, Matplotlib, Seaborn, Jupyter | [Documentation](#)

- Developed a real-time drowning detection system using YOLOv8, integrating RTSP streams and OpenCV for live video processing and rapid incident detection.
- Built and deployed the model using PyTorch, Ultralytics YOLO, and NVIDIA CUDA, with a Streamlit interface for monitoring, and utilized libraries such as torchvision, OpenCV, and HuggingFace for model performance and visualization.

Object Detection of a Rice Pest: Identification of Grasshoppers and Snails using YOLOv9 | Python, HuggingFace, streamlit, opencv-python, RTSP, ultralytics, PyTorch, torchvision, NVIDIA CUDA, TorchVision, YOLOv9, IP Camera Mobile App, NumPy, Scikit-Learn, Pandas, Matplotlib, Seaborn, Jupyter | [Documentation](#)

- Developed an object detection system for rice pests using YOLOv9 to identify grasshoppers and snails, integrated with an IP camera mobile app and deployed via Streamlit for real-time monitoring.
- Enhanced agricultural decision-making by providing automated pest detection and early warning insights, helping improve crop protection and reduce manual monitoring efforts.

Development and Prototype Implementation of a Browser Extension for Fake News Detection in Philippine News using Natural Language Processing Algorithms | Python, HuggingFace, streamlit, RoBERTa, Flask, BeautifulSoup4, Requests, PyTorch, VanillaJS, HTML, CSS, Chrome Extension API, Chrome Manifestv3, GoogleSearchAPI, NumPy, Scikit-Learn, Pandas, Matplotlib, Seaborn, Jupyter | [Documentation](#)

- Developed a functional prototype as the primary programmer for Team LiveANet's undergraduate thesis. Built a browser extension that detects fake news in Philippine news articles using NLP techniques and a RoBERTa model for real-time credibility assessment.
- Awarded Honorable Mention at the Research Colloquium for developing an innovative AI-powered fake news detection browser extension, recognized for its real-time NLP application and societal impact, and a winner of Gawad Likha 2025 (Campus Research Competition) for delivering a high-impact research solution leveraging RoBERTa and NLP techniques, demonstrating excellence in innovation, technical implementation, and practical usability.

Inquirer and Rappler Web Scraper Tool with Data Mixer | Python, Flask, requests, BeautifulSoup4, csv, os, re, pandas | [Documentation](#)

- Developed Flask-based web scrapers for Philippine news sites (Inquirer, Rappler) using Python, BeautifulSoup, and Pandas to collect, deduplicate, and merge labeled article datasets for machine learning model training and for exploratory data analysis.

SKILLS

Languages: Python, HTML, CSS, JavaScript, C++

Frameworks, Libraries & Database: PostgreSQL, Node.js, Tailwind CSS, Bootstrap, Streamlit, MySQL, TensorFlow, PyTorch, NumPy, Scikit-Learn, Pandas, Matplotlib, Seaborn, Jupyter, BeautifulSoup4

Tools: Visual Studio Code, Microsoft Word, Microsoft Excel, Microsoft Powerpoint, GitHub, Git, Jupyter, HuggingFace, Kaggle, Cursor, OpenAI, Antigravity, Linux, Kali Linux

Concepts: Robotics, AI Development, Front-End Development, Data Analysis, Data Science, Machine Learning, Cybersecurity, Networking